

INTRODUCTION

- Chronic postoperative pain affects approximately 20% of patients undergoing total knee replacement (TKR) [1].
- Recent studies have suggested that a range of factors can describe this pain. In animal studies, brain derived neurotrophic factor (BDNF) has been found to maintain central sensitization and pain amplification [2].
- BDNF is recognized as a biomarker for mood disorders [3]; its specific association with long-term postoperative pain remains largely unexplored in humans.
- Identifying factors associated with BDNF may improve understanding of chronic postoperative pain mechanisms [4].

AIM

The aim of this exploratory study was to investigate whether plasma BDNF levels correlate with clinical pain intensity, psychological factors, pain sensitivity, inflammatory markers, and microRNAs in patients five years after TKR surgery.

STUDY DESIGN & METHODS

Cohort:

- 75 knee OA patients (33 males, 42 females; age 70.85 ± 7.77 years) were evaluated five years post-TKR.

Clinical and sensory Assessments:

- Clinical outcomes included the Oxford Knee Score (OKS), Hospital Anxiety and Depression Scale (HADS), and Pain Catastrophizing Scale (PCS)
- Cuff algometry Quantitative Sensory Testing (QST) was used to assess:
- Pressure pain thresholds (PPT)
- Temporal summation of pain (TSP)
- Conditioned pain modulation (CPM).

Molecular Analysis:

- 44 inflammatory markers were measured via Luminex assay
- Next-generation sequencing detected circulating miRNAs
- Plasma BDNF concentrations were quantified using ELISA.

Statistical Analysis:

- Linear regression models with backward elimination were employed to determine if the tested markers could explain the BDNF variance

DISCLAIMERS

The authors declare no conflict of interest. CNAP is supported by the Danish National Research Foundation (DNRF121)?

RESULTS

- QST/psychological model accounted for 17.9% of variance after backward elimination ($R^2=0.179$, $P=0.002$), with HADS-depression as the only significant predictor
- miRNA model explained 61.3% ($R^2=0.613$, $P<0.001$), while inflammatory markers alone explained 83.7% ($R^2=0.837$, $P<0.001$), with IL-1a, IL-6, and VEGF as independent predictors
- Molecular model explained 86.9% ($R^2=0.869$, $P<0.001$), identifying five miRNAs and three interleukins as independent predictors.

Approach	IL-6 Model		IL-1a Model	
	Linear regression	Linear regression + Backward elimination	Linear regression	Linear regression + Backward elimination
Variable number	17	7	17	6
R ²	13.4%	27.6%	5.9%	22%

Table 1: General linear regression models with and without backwards elimination, displaying variable numbers and the adjusted R2 value.

IL-6 Linear Regression Model

Variable	Standardized Coefficient	P	R ²
Model 1			0.134
PPT. Lower leg	-0.013	0.930	
HADS depression	0.211	0.270	
hsa-miR-30d-5p	-0.299	0.219	
hsa-miR-27b-3p	-0.316	0.125	
hsa-miR-34a-5p	-0.390	0.140	
hsa-miR-4479	0.099	0.572	
hsa-miR-487b-3p	-0.091	0.573	
hsa-miR-324-5p	0.661	0.030	
VEGF	-0.043	0.811	
hsa-miR-1304-3p	-0.306	0.116	
hsa-miR-433-3p	-0.295	0.099	
hsa-miR-143-3p	0.070	0.699	
hsa-miR-217-5p	-0.026	0.899	
IL-13	-0.140	0.597	
IL-15	0.080	0.677	
IL-8	0.026	0.890	
IL-6	0.448	0.213	
Model 2			0.276
HADS depression	0.253	0.099	
hsa-miR-27b-3p	-0.348	0.034	
hsa-miR-34a-5p	-0.371	0.084	
hsa-miR-324-5p	0.448	0.017	
hsa-miR-1304-3p	-0.389	0.012	
hsa-miR-433-3p	-0.286	0.068	
IL-6	0.370	0.072	

Table 2: Linear regression model of 17 QST, pain, questionnaire factors, miRNA, inflammatory markers, and IL-6, selected to explain the variance in BDNF in patients with knee osteoarthritis 5 years after TKR. Model 1 consists of all variables. Model 2 uses backward elimination to identify independent factors explaining BDNF levels. The adjusted R² value suggests the combined predictive power of these values.

IL-1a Linear Regression Model

Variable	Standardized Coefficient	P	R ²
Model 1			0.059
PPT. Lower Leg	-0.011	0.808	
HADS depression	0.362	0.063	
hsa-miR-30d-5p	-0.179	0.495	
hsa-miR-27b-3p	-0.194	0.394	
hsa-miR-34a-5p	-0.540	0.064	
hsa-miR-4479	0.151	0.429	
hsa-miR-487b-3p	0.073	0.700	
hsa-miR-324-5p	0.543	0.072	
IL-1a	0.383	0.293	
VEGF	0.007	0.970	
hsa-miR-1304-3p	-0.399	0.060	
hsa-miR-433-3p	-0.349	0.094	
hsa-miR-143-3p	0.038	0.829	
hsa-miR-217-5p	0.013	0.944	
IL-13	-0.091	0.783	
IL-15	-0.009	0.962	
IL-8	0.115	0.525	
Model 2			0.220
HADS depression	0.298	0.063	
hsa-miR-34a-5p	-0.567	0.006	
hsa-miR-324-5p	0.526	0.004	
IL-1a	0.356	0.059	
hsa-miR-1304-3p	-0.463	0.004	
hsa-miR-433-3p	-0.432	0.013	

Table 3: Linear regression model of 17 QST, pain, questionnaire factors, miRNA, inflammatory markers, and IL-1a, selected to explain the variance in BDNF in patients with knee osteoarthritis 5 years after TKR. Model 1 consists of all variables. Model 2 uses backward elimination to identify independent factors explaining BDNF levels. The adjusted R2 value suggests the combined predictive power of these values.

CONCLUSIONS

This study provides preliminary evidence that BDNF serves as a potential biomarker for a range of inflammatory and epigenetic biomarkers that are associated with chronic pain five years after TKR.

While these exploratory results are promising, external validation is necessary to confirm BDNF's utility in clinical practice

REFERENCES

- Hertel et al., 2024 – PAIN; Quantitative sensory testing, psychological factors, and quality of life as predictors of current and future pain in patients with knee osteoarthritis.
- Xiong et al., 2024 – Biomolecules; The Role of the Brain-Derived Neurotrophic Factor in Chronic Pain: Links to Central Sensitization and Neuroinflammation
- Cavaleri et al., 2023 – Neuroscience & Biobehavioral Reviews; The role of BDNF in major depressive disorder, related clinical features, and antidepressant treatment: Insight from meta-analyses
- Sorkpor et al., 2021 – Biol Res Nurs; The Relationship Between Plasma BDNF and Pain in Older Adults With Knee Osteoarthritis

